

LECTURE 3: Independence

- Independence of two events
- Conditional independence
- Independence of a collection of events
- Pairwise independence
- Reliability
- The king's sibling puzzle

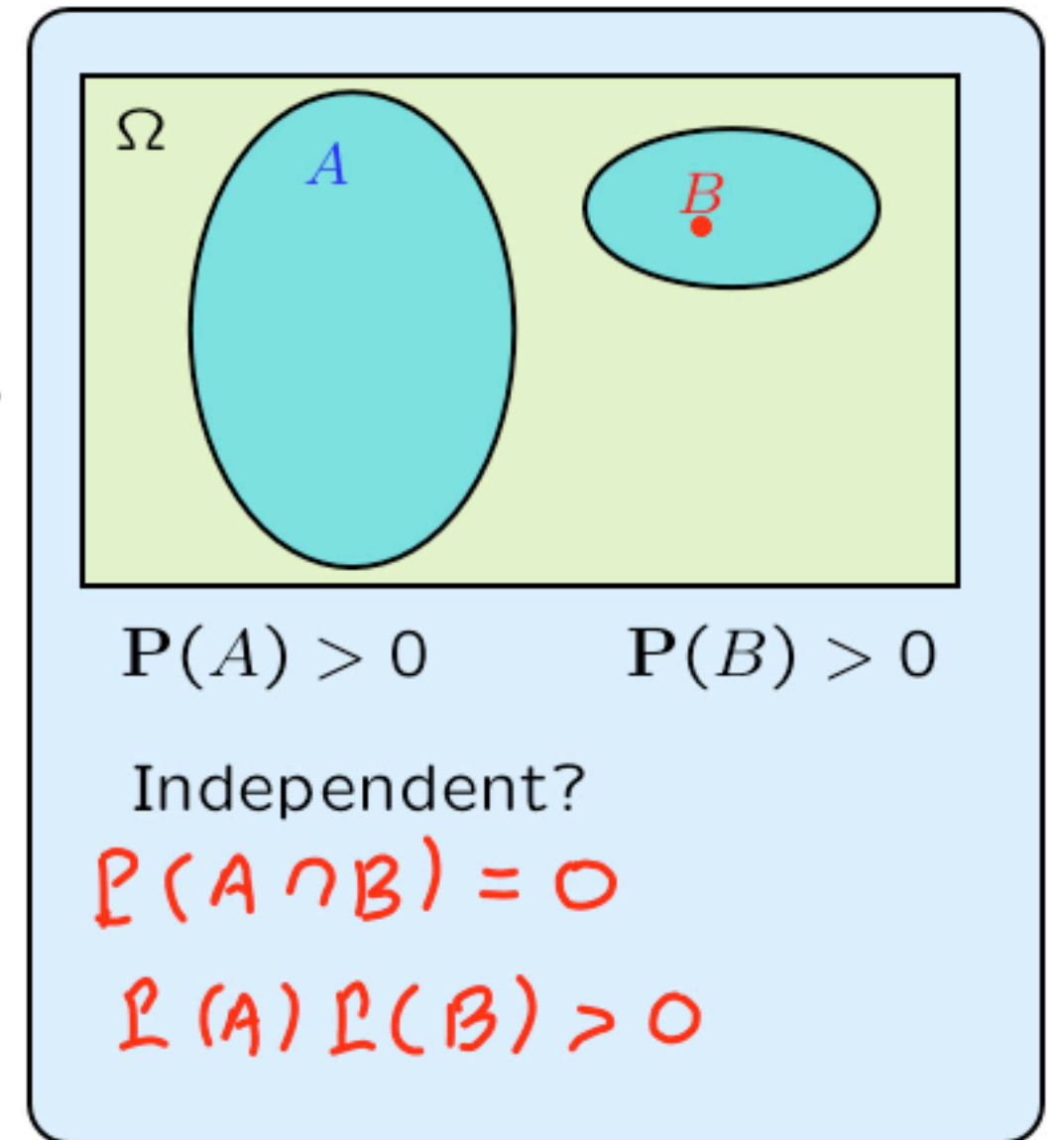
Independence of two events

- Intuitive “definition”: $P(B | A) = P(B)$
 - occurrence of A provides no new information about B

$$\begin{aligned} P(A \cap B) &= P(A) P(B|A) = P(A) P(B) \\ &= P(B) P(A|B) = P(B) P(A) \end{aligned}$$

Definition of independence: $P(A \cap B) = P(A) \cdot P(B)$

- Symmetric with respect to A and B
- implies $P(A | B) = P(A)$
- applies even if $P(A) = 0$



$$\text{Se } P(A) = 0$$

$$\downarrow. A \cap B \subseteq A \Rightarrow P(A \cap B) \leq P(A) \Rightarrow P(A \cap B) = 0$$

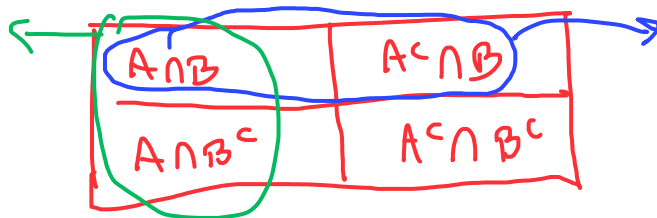
$$\Rightarrow P(A \cap B) = P(A)P(B) \text{ quando } P(A) = 0$$

Exemplo: A: eu acordei antes das 6:00
 B: dolar às 17:20 vale mais
 que 5,20

$$P(A) = 1/2$$

$$P(B) = 0,9$$

$$\begin{aligned} P(A) &= 0,5 \\ &= P(A \cap B) + \\ &\quad P(A \cap B^c) \end{aligned}$$



$$\begin{aligned} P(B) &= 0,9 \\ &= P(A \cap B) + \\ &\quad P(A^c \cap B) \end{aligned}$$

Para independência, $P(B|A) = 0,9$

$$= \frac{P(B \cap A)}{P(A)} = \frac{P(B \cap A)}{0,5} \Rightarrow P(B \cap A) = 0,45$$

De forma parecida, podemos calcular $P(B \cap A^c)$, $P(A \cap B^c)$, $P(A \cap B)$

Independence of event complements

$$P(A \cap B^c) \stackrel{?}{=} P(A)P(B^c)$$

Definition of independence: $P(A \cap B) = P(A) \cdot P(B)$

- If A and B are independent, then A and B^c are independent.

– Intuitive argument

– Formal proof



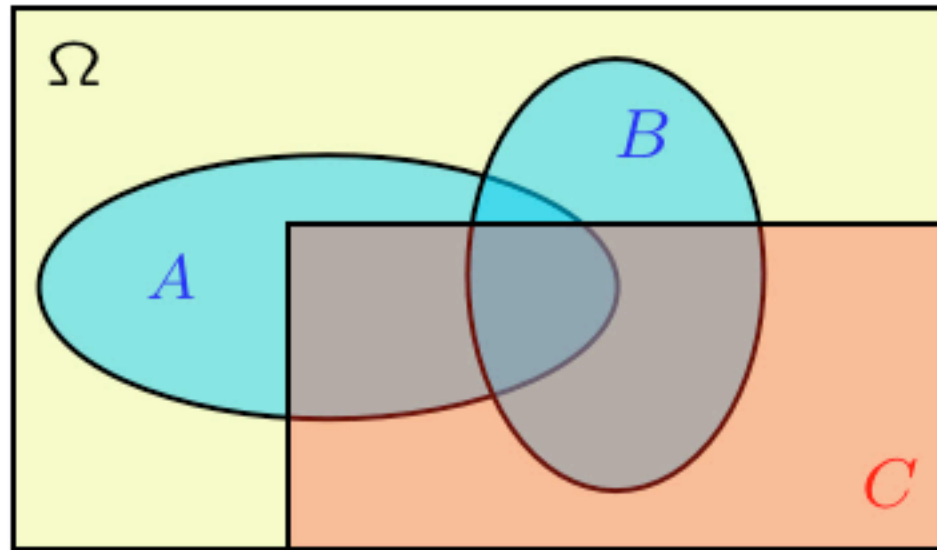
$$A = (A \cap B) \cup (A \cap B^c)$$

$$\begin{aligned} P(A) &= P(A \cap B) + P(A \cap B^c) \\ &= P(A)P(B) + P(A \cap B^c) \end{aligned}$$

$$\begin{aligned} P(A \cap B^c) &= P(A) - P(A)P(B) = P(A)(1 - P(B)) \\ &= P(A)P(B^c) \end{aligned}$$

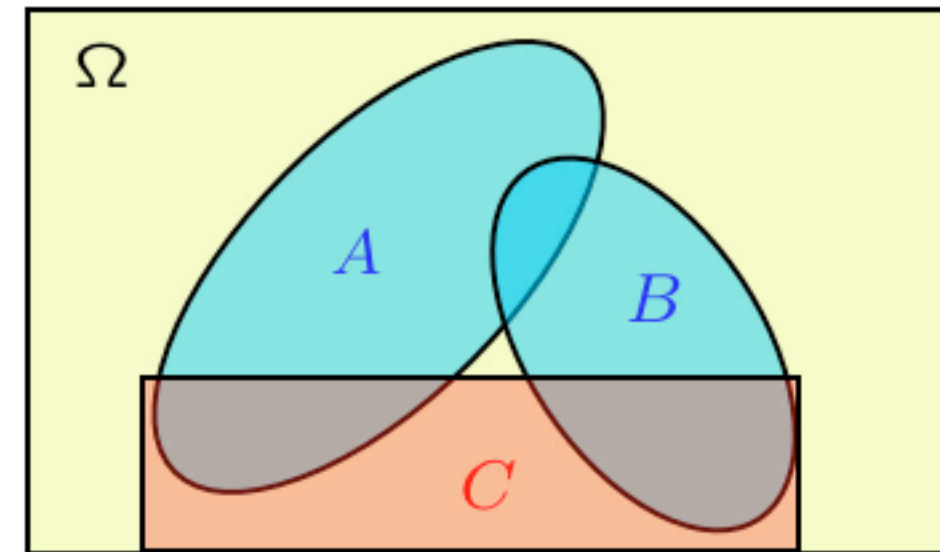
Conditional independence

- Conditional independence, given C , is defined as independence under the probability law $\mathbf{P}(\cdot | C)$

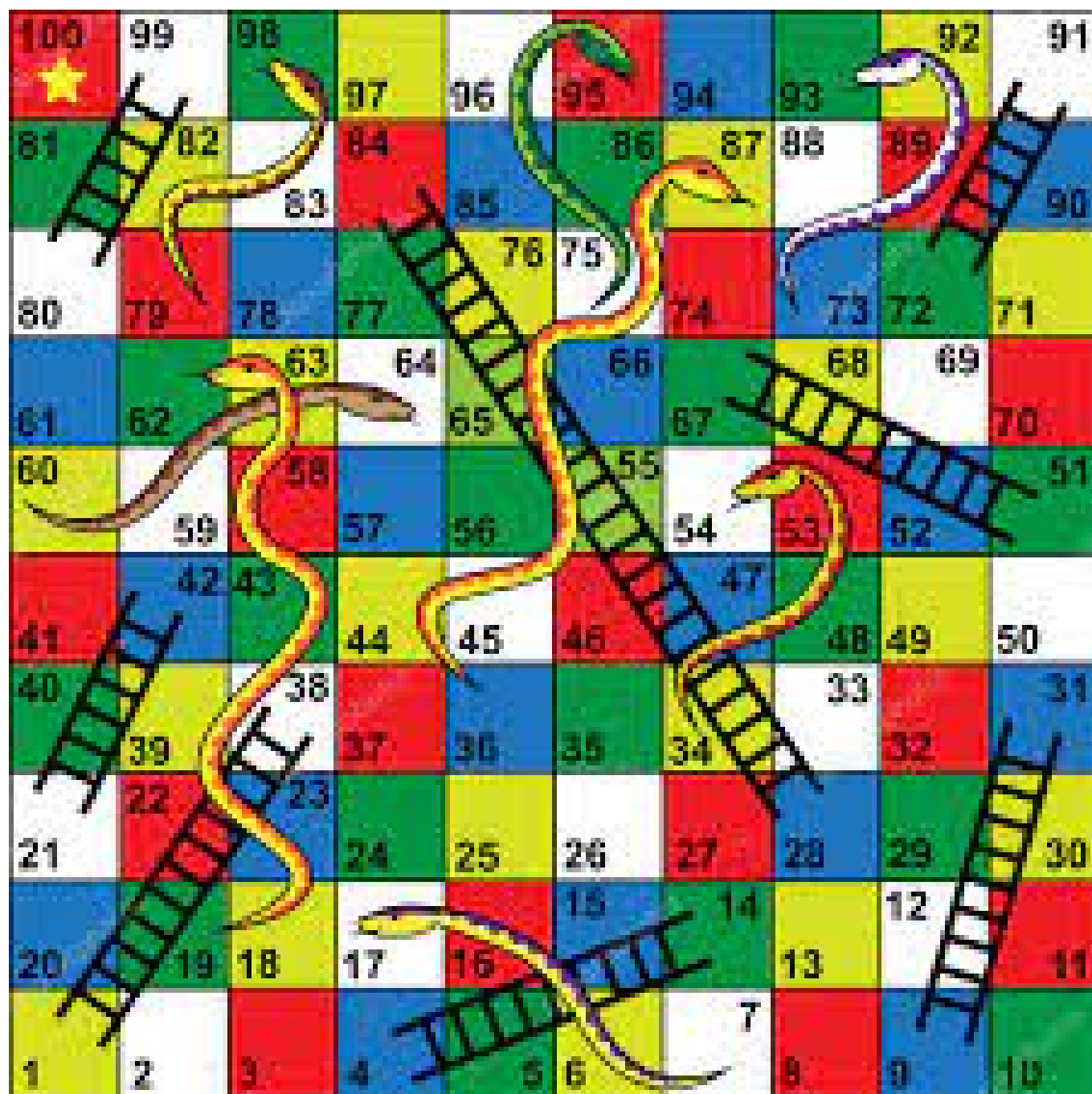


$$P(A \cap B | C) = P(A | C) P(B | C)$$

Assume A and B are independent



- If we are told that C occurred, are A and B independent? **No**



$P(\text{Casa 21, rodada 3} \mid$
 $\text{Casa 14, rodada 1}) \neq$

$P(\text{Casa 21, rodada 3})$

Por outro lado,

$P(\text{Casa 21, rodada 3} \cap$
 $\text{Casa 14, rodada 1} \cap$
 $\text{Casa 17, rodada 2}) =$

$P(c. 21, r. 3 \mid c. 17, r. 2) \times$

$P(c. 14, r. 1 \mid c. 17, r. 2)$

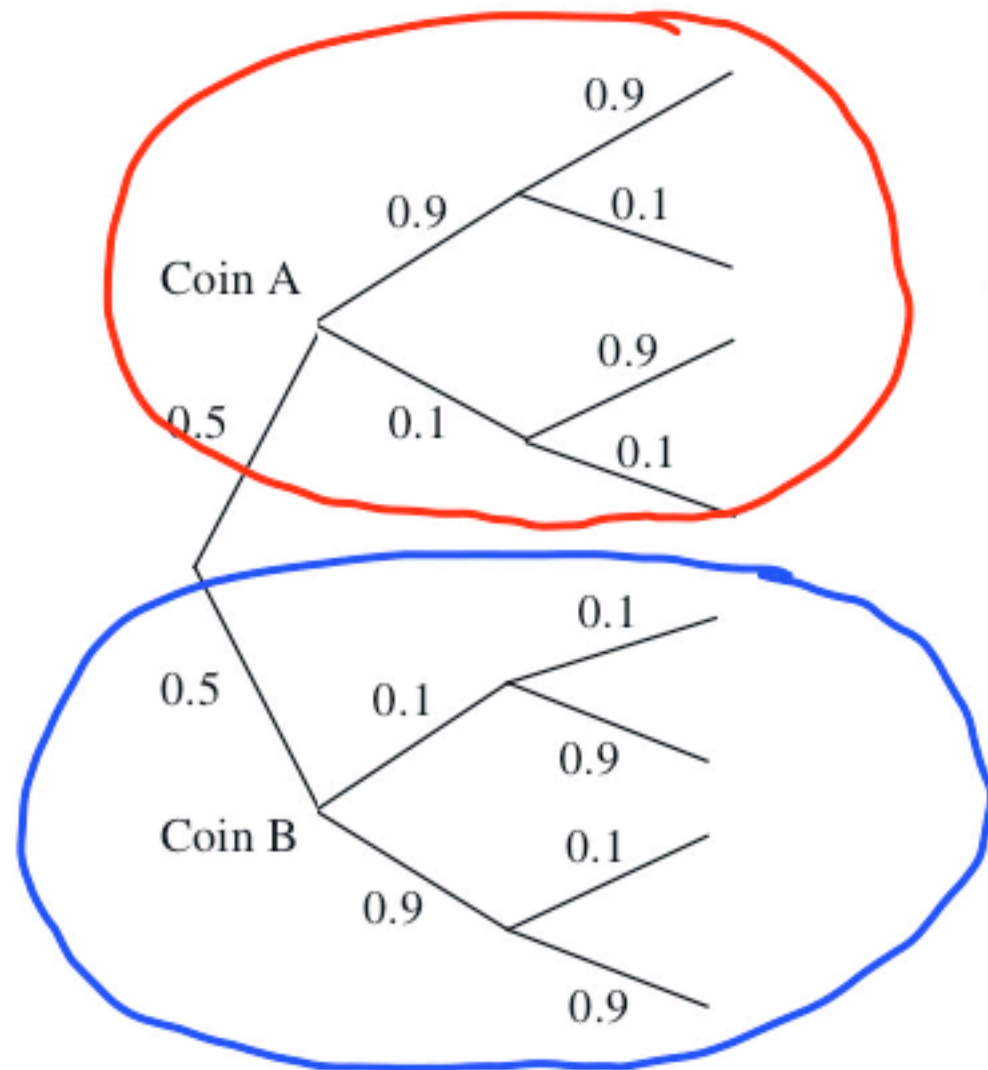
Conditioning may affect independence

- Two unfair coins, A and B :
 $P(H \mid \text{coin } A) = 0.9$, $P(H \mid \text{coin } B) = 0.1$
- choose either coin with equal probability

given a coin:
independent tosses

- Are coin tosses independent?

No!



– Compare:

$$P(\text{toss } 11 = H) = P(A)P(H_{11}|A) + P(B)P(H_{11}|B) \\ = 0.5 \times 0.9 + 0.5 \times 0.1 = 0.5$$

$$P(\text{toss } 11 = H \mid \text{first 10 tosses are heads})$$

$$\approx P(H_{11} \mid A) = 0.9$$

Seja $H_{10} = \{\text{cara nos 10 primeiros lançamentos}\}$

$$P(H_{11} | H_{10})$$

Prob. total: $P(H_{11}) = P(A)P(H_{11}|A) + P(B)P(H_{11}|B) \Rightarrow$ no mundo original

$$P(H_{11} | H_{10}) = P(A | H_{10}) P(H_{11} | A, H_{10}) + P(B | H_{10}) P(H_{11} | B, H_{10}) \Rightarrow \text{no mundo de } H_{10}$$

$$\frac{P(H_{10} | A) P(A)}{P(H_{10})} = P(B) P(H_{10} | B) + P(A) P(H_{10} | A)$$

$\nearrow 0,9^{10}$ $\nearrow 0,3$

"
0,9 por indep.
condicional

$P(H_{11} | B)$ por independência
" condicional
0,1

$\nwarrow 0,1^{10}$

$\searrow 0,9^{10}$

Independence of a collection of events

- **Intuitive “definition”:** Information on some of the events does not change probabilities related to the remaining events

$$A_1, A_2, \dots, \text{indep} \Rightarrow P(A_3 \cap A_4^c) = P(A_3 \cap A_4^c | A_1 \cup (A_2 \cap A_5^c)) \quad .$$
$$P(A_3) = P(A_3 | A_1 \cap A_2) = P(A_3 | A_1 \cap A_2^c) = P(A_3 | A_1^c \cap A_2)$$

Definition: Events $A_1, A_2, \dots, A_n$ are called **independent** if:

$$P(A_i \cap A_j \cap \dots \cap A_m) = P(A_i)P(A_j) \dots P(A_m) \quad \text{for any distinct indices } i, j, \dots, m$$

$n = 3$:

$$\left. \begin{aligned} P(A_1 \cap A_2) &= P(A_1) \cdot P(A_2) \\ P(A_1 \cap A_3) &= P(A_1) \cdot P(A_3) \\ P(A_2 \cap A_3) &= P(A_2) \cdot P(A_3) \end{aligned} \right\} \text{ pairwise independence}$$

$$P(A_1 \cap A_2 \cap A_3) = P(A_1) \cdot P(A_2) \cdot P(A_3)$$

Independência par a par

x, y : bits $v = x \oplus y$ (códigos corretores, criptografia)

$$\Omega = \{(0, 0, 0), (0, 1, 1), (1, 0, 1), (1, 1, 0)\}$$

$$A = \{x = 1\}$$

$$B = \{y = 0\}$$

$$C = \{v = 1\}$$

$$P(A) = P((1, 0, 1) \cup (1, 1, 0)) = \frac{1}{2} = P(B) = P(C)$$

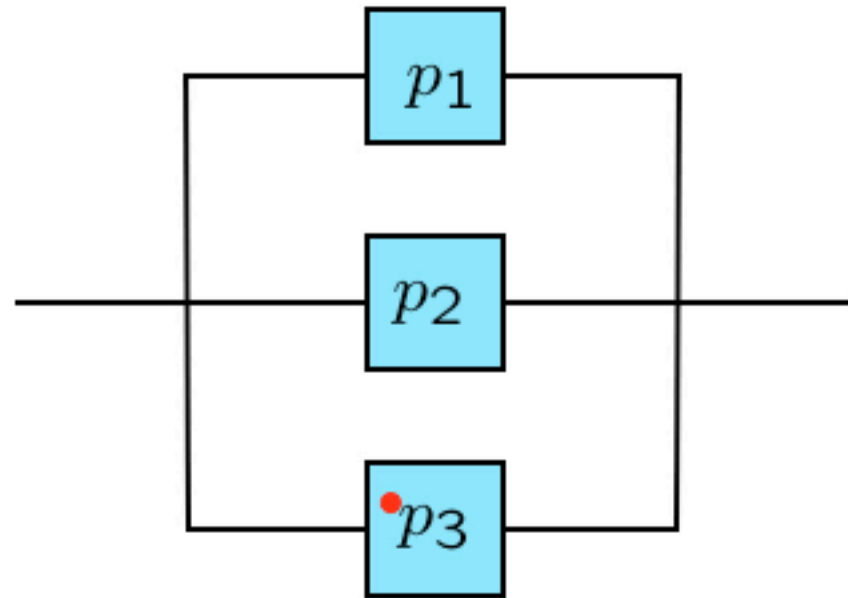
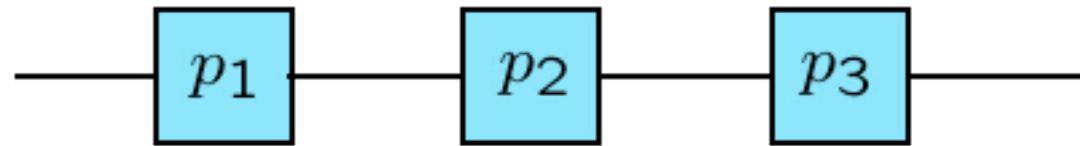
$$P(C \cap A) = P((1, 0, 1)) = \frac{1}{4} = P(A) P(C)$$

$$P(C | A, B) = 1 \neq P(C)$$

Reliability

p_i : probability that unit i is "up"

independent units



U_i : i th unit up

$U_1, U_2, \dots, U_n$ independent

F_i : i th unit down

$\Rightarrow F_i$ independent

probability that system is "up"?

$$P(\text{system up}) = P(U_1 \cap U_2 \cap U_3)$$

$$= P(U_1) P(U_2) P(U_3) = p_1 p_2 p_3$$

$$P(\text{system is up}) = P(U_1 \cup U_2 \cup U_3)$$

$$= 1 - P(F_1 \cap F_2 \cap F_3)$$

$$= 1 - P(F_1) P(F_2) P(F_3)$$

$$= 1 - (1 - p_1)(1 - p_2)(1 - p_3)$$